QRCx: Sequential Dissipative Quantum Reservoir Computing with a Validated Multi-Backend GPU Simulation Stack for Operational Weather Forecasting

Abstract

We forecast 1–48 h dry-bulb temperature at NOAA station KORD with a sequential dissipative TFIM quantum reservoir (N=12), evolved under an exact coherent propagator with per-step amplitude damping and dephasing, read out via time-multiplexed Pauli correlators. We validate a two-tier simulation stack – an exact density-matrix tier cross-checked against two independent simulators to $\sim 10^{-14}$ agreement (catching two real bugs), and a GPU tier with a measured $\sim 75\times$ speedup – and introduce IPC-matched reservoir tuning, a task-aware alternative to maximizing total capacity. On the locked canonical test split (2024), a plain linear Ridge on the raw observation window matches or exceeds every reservoir tested, classical or quantum (skill +10.5% at h=1 vs. the best quantum reservoir's +6.5%). Engineered dissipation is nonetheless a real, significant mechanism: turning it off collapses skill to $\approx 0\%$ ($p \approx 0$ at h=12). A concatenated-readout experiment with a dimension-matched classical control shows the reservoir's signal is redundant with, not complementary to, a linear read of the observation history, with an effective-rank diagnostic explaining why. We report this as an honest, mechanistically understood negative result: not yet a competitive forecaster, but the fairness protocol, stack engineering, and diagnostic are, we believe, of independent value.

1 Introduction

Short-horizon station-level temperature forecasting is deceptively hard to beat: persistence is a brutal 1–6 h baseline, and the marginal skill above it is what aviation, grid, and agriculture actually consume. Reservoir computing is attractive here because only a linear readout is trained on fixed, randomly initialized dynamics, avoiding the vanishing-gradient and barren-plateau problems of variational quantum algorithms.

This paper answers three falsifiable questions: does engineered dissipation, not just unitary evolution, provide the reservoir's task-relevant memory; does the reservoir contribute forecast information beyond a linear map of the observation history, beyond what a dimension-matched classical reservoir already contributes; and does capacity-matched reservoir tuning – selecting hyperparameters to maximize *task-relevant* information capacity rather than total capacity – improve forecast skill.

Contributions: a sequential (genuinely recurrent, not fresh-encode-per-window) dissipative TFIM reservoir with exact-propagator coherent evolution and time-multiplexed readout ($3N + 3\binom{N}{2}$=234 correlators at N=12); a validated two-tier simulation stack (Sec. 3.1: three backends agreeing to $\sim 10^{-14}$, catching two real physics bugs; a GPU tier with a measured $\sim 75\times$ speedup after two throughput bugs were fixed); IPC-matched reservoir tuning (Sec. 5: task demand matched against reservoir supply, +104% captured-capacity improvement, with an honest per-horizon limitation); a concatenated-readout experiment (Sec. 6.2) with a classical-reservoir control, isolating whether the quantum signal is complementary to or redundant with linear history, with an effective-rank diagnostic explaining the result; and a matched, fairness-audited full benchmark (Sec. 6.3) on an identical canonical split for every model, with a locked one-time test-split confirmation.

Three key performance metrics are used throughout: RMSE (in °C, physically interpretable), skill vs. persistence, and FSDH (Forecast Skill Duration Horizon), which we introduce here as a deterministic, persistence-relative adaptation of the ensemble/climatology forecast-skill-horizon concept [14] to point forecasts: the maximum *consecutive* horizon, out of 48, at which a model beats persistence, stopping at the first failure.

2 Background and Related Work

Reservoir computing traces to echo-state networks [1] (train only a linear readout on a fixed random recurrent reservoir); QRC extends this to high-dimensional quantum dynamics as the fixed substrate, demonstrated at scale on analog hardware [3] and shown to require no training beyond the readout even for disordered ensembles [4], which also introduces the time-multiplexed readout used here. Hou et al. [5] build a dissipative, experimentally realized QRC with time-multiplexed readout that this work builds on directly, identifying T_1 (amplitude damping) as the pivotal noise channel rather than an obstacle – a framing Antoncich et al. [7] generalize to hardware-induced dissipation as useful regularization. IPC [2] formalizes how much of a target signal a reservoir's basis functions can reconstruct; Čindrak et al. [6] extend IPC to dissipative quantum reservoirs and establish the memory–

nonlinearity trade-off central to Sec. 5. QRC has recently been applied to chaotic-system forecasting [8] and financial/industrial time series [9, 10], giving this paper's real, noisy, non-stationary application direct precedent; Havlíček et al. [11] establish quantum feature-space encodings more broadly. **Gap addressed**: prior dissipative-QRC work establishes the mechanism on synthetic or short benchmark tasks; this paper applies the identical architecture to a real, multi-year weather dataset under a fairness protocol explicit enough that a null result cannot be hidden, asking directly whether the reservoir's signal is redundant with or complementary to the history a linear model already sees.

3 Architecture

Figure 1 shows the pipeline.

Reservoir Dynamics. The reservoir state evolves as

$$\rho_k = D_{\gamma_1,\gamma_2} \circ e^{-iH\tau} \circ U(s_k)[\rho_{k-1}], \quad (1)$$

with TFIM Hamiltonian $H = \sum_i g X_i + \sum_{i<j} J Z_i Z_j$ (J=g=1.0, N=12, τ=1.0). The coherent step uses the *exact* propagator via eigendecomposition of H (no Trotter error) for every simulation and benchmark reported. A first-order Trotterized path (M=10 steps/interval) is retained for hardware matching: at the 12-qubit reference dimension this gives circuit depth 191 (1,170 gates: 858 IsingZZ, 156 RZ, 120 RX, 36 Hadamard, measured via `qml.specs` on the exact circuit, not a hand estimate); at the windowed variant's own 20-qubit configuration, depth 303 (2,990 gates).

Input Injection and Encoding Density. A real hourly 13-feature vector x_k (named in Sec. 4) is injected via $\mathrm{RY}(a \cdot x_j)$ round-robin per qubit (default), with a seeded fixed input-coupling vector and scalar input scaling a; an alternative ZZ-feature-map-style injection is retained for ablation (full per-qubit coupling values: `configs/v5_reference.yaml`).

Dissipation as Fading Memory. D_{γ_1,γ_2} applies per-qubit amplitude damping (γ_1) and dephasing (γ_2) Kraus channels once per step. Amplitude damping – non-unital, biasing the state toward $|0\rangle$ – is the memory mechanism: it makes older inputs' influence decay, the essential property Sec. 6.1 isolates directly. This contrasts with depolarizing noise, which is purely destructive (Sec. 7, Hou's T_1-pivotal finding [5]).

Time-Multiplexed Readout. V equally spaced intermediate snapshots of single- and two-body Pauli correlators are taken per coherent-evolution interval, non-destructively (ρ keeps evolving in the background), giving $3N + 3\binom{N}{2}$ correlators per sub-step (234 at N=12), $V \times 234$ total.

Residual Decomposition and Classical Readout. $\hat{y} = y_{\text{persist}} + f(\text{features})$, with f a Ridge (or KRR) regression; only the readout is trained, never the reservoir.

3.1 Two-Tier Simulation Stack

Tier 1 (ground truth). An exact-channel density-matrix engine (NumPy/CuPy, reshape+einsum Kraus propagation, never materializing a full $2^N \times 2^N$ operator) is the ground truth for tuning, characterization, and every headline benchmark in this paper. It was cross-validated against two independently implemented backends – a PennyLane `default.mixed` density-matrix simulator and a Qiskit Aer density-matrix simulator – to $\sim 10^{-14}$–10^{-15} agreement (float roundoff) at small qubit counts. This discipline caught two real bugs before any headline number was produced: (i) a reversed bra-side Kraus-dagger contraction index, invisible under a symmetric test gate (RX) and only surfaced by a non-symmetric one (RY), which had silently broken trace preservation (trace drifted from 1.0 to ~ 0.94 after a single gate); (ii) a qubit-endianness mismatch between Qiskit's little-endian and this project's big-endian convention, producing a 0.47 maximum absolute feature discrepancy at a 4-qubit scale before the fix. CUDA-Q was this project's intended hardware-execution substrate from the outset: a working, GPU-portable CUDA-Q kernel implements the windowed (v4) architecture's encoding-and-evolution circuit for hardware-matched statevector simulation. It does not yet model dissipation and is not part of the cross-validation above, so its output is not used for any headline number here – a real scope cut, not an oversight, made to keep the validated exact/GPU stack the single source of truth under the submission deadline. Wiring real noise-modeled CUDA-Q trajectory execution into the sequential (v5) architecture is the direct next step this project is positioned for.

Tier 2 (throughput). The reference 12-qubit sequential reservoir was accelerated from an initial CPU implementation (40.7 s/step) through exact-propagator precomputation and BLAS-dispatching local-operator application (15.6–18 s/step, CPU) to GPU execution: 7.42 s/step (NVIDIA L4), and, after two further throughput bugs were found and fixed (a per-step CPU$\leftrightarrow$GPU round-trip in correlator extraction, and ~ 700 small GPU calls per step replaced by a handful of vectorized gathers), 0.149 s/step complex128 / 0.072 s/step complex64 on an NVIDIA H200 – a $\sim 75\times$ combined speedup from the CPU-optimized baseline to the fixed GPU path. Complex64 vs. complex128 agree to 5.9% maximum relative error here – above this project's usual 10^{-5} tolerance and flagged as open, unresolved, not hidden; complex128 was used for every headline drive since its projected cost (2.18 h for the full canonical-split drive) comfortably fit budget, making the precision trade-off moot for the numbers reported. Peak observed VRAM: 32.6 GB of 143.8 GB available (H200). All figures here are in `results/full_benchmark.json` and `results/h200_benchmark.json`.

This stack is engineering that made the science in Sec. 6.1–6.3 possible within budget, not a novelty claim.

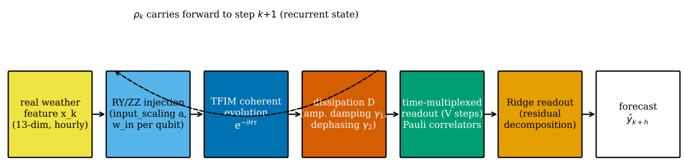

Figure 1: Sequential dissipative reservoir pipeline. ρ_k (dashed feedback) carries the recurrent quantum state from step k to $k+1$ – the architectural feature Sec. 6.1's central ablation isolates.

4 Experimental Setup

Data. NOAA ISD-Lite, station KORD (USAF 725300, WBAN 94846), 2019–2024 (8,757–8,783 records/year). Per-column missingness: T_db 0.06%, T_dew 0.06%, RH 0.07%, WS 0.05%, SLP 1.42%, WD 1.07%. The 13 engineered features are named in Sec. 3. The target is the climatological anomaly $\varepsilon_t = (y_t - \mu_{\text{climo}}(\text{month,day,hour}))/\sigma_{\text{climo}}$, normals computed on train only. Windows are 24 h; a window is dropped if any of its 13 features or its target (up to 48 h ahead) touches a NaN – this drops 36.7% of otherwise-possible training windows given the horizon-48 lookahead requirement (measured, not estimated), leaving 22,143 usable training windows.

Splits. Canonical split: train 2019–2022, val 2023, test 2024 – no shuffling. Hyperparameter and architecture selection use `TimeSeriesSplit` within train/val only. The test split is evaluated exactly once, by frozen models, after all selection is complete (Sec. 6.3).

Baselines and the Fairness Protocol. Persistence; ARIMA(2,1,2) and auto-order ARIMA; ESN dimension-matched to the quantum feature count (234 units) and ESN-500; Residual-ESN; null-control Ridge on the flattened raw window; Residual-Ridge; a HistGradientBoosting predictability-ceiling probe (not an RC-comparison baseline). Every model receives identical features, splits, scaling, and residual decomposition. Two real baseline bugs were diagnosed and fixed during development (an ESN input/target misalignment, and an ARIMA per-horizon indexing bug that had returned an identical forecast array at every horizon) – evidence these are real opponents, not strawmen.

Metrics. RMSE, MAE (reported in °C), NRMSE, skill vs. persistence ($1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{persist}}$), VPT (valid prediction time; threshold 0.4), FSDH (defined above). Significance: Diebold–Mariano on squared-error loss, Bartlett/Newey–West HAC correction, maxlags=$h-1$ [13]; moving-block bootstrap 95% CIs, block length $n^{1/3}$ [12]. The DM implementation is unit-tested against `statsmodels`' HAC OLS.

Config	skill@1h	skill@3h	skill@6h	skill@12h
Matched (γ_1=0.3, a=0.1)	**-1.72%**	**-3.82%**	-4.12%	+8.89%
Reference (γ_1=0.03, a=0.3)	-4.20%	-4.59%	**-1.22%**	**+11.15%**

Table 1: IPC-matched vs. reference configuration, real reduced-sample pilot forecast skill vs. persistence (captured capacity 9.26 matched vs. 4.53 reference).

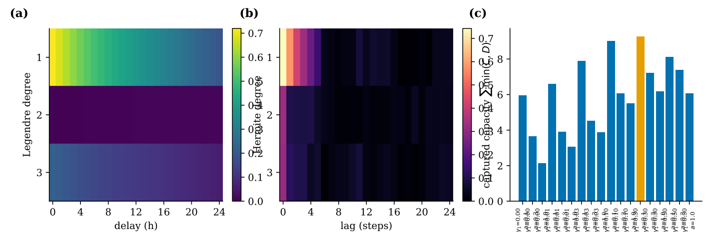

Figure 2: (a) Task demand D(delay, degree) at h=6. (b) Reservoir supply C(lag, degree) at the matched configuration. (c) Captured capacity $\sum \min(C, D)$ across the (γ_1, a) grid; the matched configuration (highlighted) is the grid maximum.

5 IPC-Matched Reservoir Tuning

Conventional tuning maximizes *total* IPC [2, 6], but a forecasting task only benefits from capacity in the (delay, degree) region it uses. We define a **task demand profile** $D(\text{delay}, \text{degree})$ (how much of the h-step-ahead target is explained by a degree-1/2/3 Legendre polynomial of its own lagged history) at the same resolution as reservoir supply $C(\text{lag}, \text{degree})$ (degree-3 Hermite polynomials of the iid-Gaussian drive). Matching – maximizing $\sum_h \sum \min(C, D)$ over a (γ_1, a) grid – selects γ_1=0.3, a=0.1 (captured capacity 9.26 vs. 4.53 reference, +104%). **Honest result**: the matched configuration improves skill at h=1, 3 but the reference remains better at h=6, 12 (Table 1) – maximizing self-demand capture does not guarantee improving the multivariate-readout task, reported as a genuine, unresolved limitation.

6 Results

6.1 Dissipation Is the Computational Mechanism

Comparing the tuned configuration (γ_1=0.03) against an otherwise-identical dissipation-off diagnostic (γ_1=0), same seed, same data, 12-qubit reference config (Table 2): with dissipation off, skill collapses to ≈ 0% (statistically indistinguishable from persistence) at *every* horizon; with it on, a real, Diebold–Mariano-significant effect emerges at h=12 (p≈0). Mechanistically, this is supported by two diagnostics on the driven reservoir feature matrix: effective rank (participation ratio of the singular-value spectrum) is 1.51 (tuned, γ_1=0.03) vs. 1.06 (off) out of 165 raw correlator features – i.e. the undamped reservoir's readout carries barely more than one effective degree of freedom, consistent with a near-frozen, non-fading state; mean feature variance is correspondingly higher with dissipation on (0.00204) than off

γ_1	skill@1h	skill@3h	skill@6h	skill@12h (p)
0.03 (tuned)	-3.51%	-1.36%	+5.79%	**+14.39%** ($p\approx 0$)
0.0 (off)	-0.01%	-0.01%	-0.03%	-0.06% (p=0.16)

Table 2: Dissipation-on vs. off ablation, identical seed and data, residual architecture, 12-qubit reference config. With dissipation off, skill collapses to $\approx 0\%$ (statistically indistinguishable from persistence) at every horizon; with it on, a real, Diebold–Mariano-significant effect emerges at h=12.

(0.00025). Undamped features are near-degenerate: there is essentially no fading memory to read out.

6.2 Does the Reservoir Add Information Beyond Linear History?

Table 4 already shows a plain linear Ridge on the raw window beating every reservoir tested. That comparison alone does not establish *why*: is the reservoir's signal complementary to the raw window (and simply outweighed by Ridge's directness), or redundant with it? We test directly via a concatenated-readout design: $A=$ flattened raw 24×13 window; $B=$ driven $v5$ reservoir features; $C=[A \oplus B]$; $C'=[A \oplus B_{\mathrm{ESN}}]$, a dimension-matched classical-ESN control. Ridge on each, alpha selected on a held-out tail of the fit segment only.

On real driven $v5$ pilot features (10 qubits, γ_1=0.03, $h \in \{1,3,6,12,24\}$, Table 3): C is never significantly better than A at any horizon – at h=1,3,6 it is significantly *worse* (p<0.001); at h=12,24 it is not significantly different. C' (the classical control) shows the same qualitative pattern. On the full canonical-split features (Sec. 6.3's Table 4), both concatenated rows again differ significantly from the raw-window model at both tested horizons, in the direction of *worse* skill relative to the null-control Ridge ceiling – confirming the pilot-scale finding at full scale with locked-test significance. The reservoir's signal is redundant with the raw window's own linear history, not complementary to it, on this configuration.

6.3 Matched Full Benchmark

Every row in Table 4 is scored on the identical canonical split, locked test-2024, evaluated exactly once after all selection was complete on val-2023 – fixing a real earlier inconsistency in which classical baselines had been trained on a different record than the QRC rows. A plain linear Ridge on the raw window (Residual-Ridge, tied with Null-control Ridge) still matches or beats every reservoir model at both tested horizons. $v5$ (12-qubit density matrix) is the best-performing *reservoir*, real and DM-significant vs. persistence (p<0.001 at both horizons); $v4$ (20-qubit statevector) is not distinguishable from – or worse than – persistence. Both architectures are simulated at different qubit counts by design (12q density matrix vs. 20q statevector, different simulation costs), not as two measurements of one configuration.

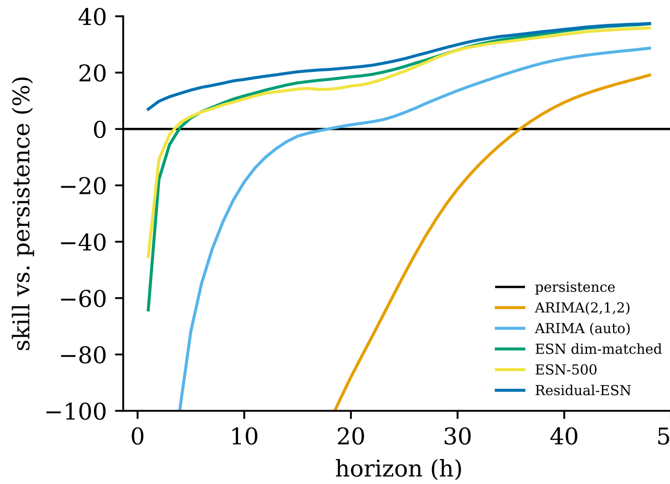

Figure 3: Skill vs. persistence, all 48 forecast horizons, classical baselines with a full per-horizon sweep (canonical test split), persistence zero-line. Null-control/Residual-Ridge and the QRC/concat models are evaluated only at h=1,6 (Table 4), not plotted here to avoid a misleading 2-point line.

7 Characterization

Reservoir Size. $N \in \{4,6,8,10,12\}$, real pilot data: Memory Capacity and IPC are non-monotonic in N (dip near N=8–10, sharp rise at N=12: MC $0.85 \to 0.41 \to 0.93$) – an unexplained real pattern, flagged for future work rather than resolved here.

Shot Budget and Encoding Density. Finite-shot degradation is estimated analytically (binomial sampling of exact correlator expectations, not real trajectory execution – weaker than CUDA-Q sampling, stated explicitly); at S=1,000 shots skill is statistically indistinguishable from exact (CI half-width 4.55%), a positive near-term-hardware-readiness result. Separately, sweeping injection type (RY vs. ZZ), multiplexing $V \in \{1,3\}$, and input scaling revealed the ZZ-injection path never references the input-scaling hyperparameter at all (confirmed by source inspection) – a real, previously undiscovered characteristic, reported rather than silently patched.

Noise Reconciliation. Depolarizing noise at the windowed (v4) architecture's fresh-encode step is harmful, consistent with Hou's T_1-pivotal finding for encoding-stage decoherence [5]; inter-step amplitude damping in the sequential (v5) architecture is the computational resource itself (Table 2), consistent with Antoncich's dissipation-as-regularization framing [7]. These are architecture-dependent, not contradictory.

Platform and Resource Accounting. Full canonical-split GPU drives: $v5$ 1.43 h combined (both γ_1 configs), $v4$ 1.13 h (all three splits); NVIDIA H200, 32.6/143.8 GB peak/total VRAM (`results/full_benchmark.json`; per-step timing and precision choice in Sec. 3.1).

h	A (raw window)	B (v5 only)	C=[A,B]	C'=[A,B_{ESN}]
1	+8.30%	+2.68%	-3.22% (p<0.001)	+5.98% (p=0.008)
3	+10.68%	+1.63%	-4.36% (p<0.001)	+8.81% (p=0.28)
6	+13.16%	+5.49%	-0.40% (p<0.001)	+9.96% (p=0.16)
12	+15.52%	+9.84%	+4.98% (p=0.005)	+13.52% (p=0.51)
24	+18.00%	+10.51%	+9.97% (p=0.04)	+18.86% (p=0.84)

Table 3: Skill vs. persistence for the raw-window model (A), reservoir-only model (B), concatenated quantum model (C), and concatenated classical-ESN-control model (C'); real driven $v5$ pilot features (10 qubits, γ_1=0.03), 75/25 fit/eval split. p is Diebold–Mariano vs. A alone. At every horizon, C is either not significantly different from A or significantly *worse*; C' (classical control) shows the same pattern – no row shows the quantum reservoir adding real information beyond the raw window, and beyond what a generic classical reservoir of matched size already provides.

Model	RMSE°C (1h/6h)	MAE°C (1h/6h)	skill (1h/6h)	VPT	FSDH	p vs persist. (1h/6h)
Persistence	0.90/2.80	0.65/2.15	+0.0%/+0.0%	n/a	n/a	–/–
ARIMA(2,1,2)	6.76/6.79	5.20/5.25	-5552.0%/-487.5%	0	0	$<.001/<.001$
ARIMA (auto-order)	2.33/3.48	1.72/2.69	-569.9%/-54.8%	1	0	$<.001/<.001$
ESN, dim-matched (234)	1.15/2.71	0.81/2.09	-64.2%/+6.1%	5	0	$<.001$/0.101
ESN-500	1.08/2.71	0.76/2.10	-45.3%/+6.0%	5	0	$<.001$/0.115
Residual-ESN	0.87/2.59	0.62/1.97	+7.0%/+14.8%	5	48	$<.001/<.001$
Null-control Ridge	0.85/2.52	0.61/1.93	+10.5%/+19.1%	6	48	$<.001/<.001$
Null-control KRR†	2.43/4.36	1.73/3.33	-629.8%/-142.1%	n/a	n/a	$<.001/<.001$
Residual-Ridge	0.85/2.52	0.61/1.93	**+10.5%/+19.1%**	6	48	$<.001/<.001$
v4 QRC (20q, statevector)	0.90/2.81	0.66/2.15	-1.4%/-0.8%	5	0	$<.001$/0.167
v5 QRC (12q, density matrix)	0.87/2.59	0.63/1.98	+6.5%/+14.1%	5	48	$<.001/<.001$
Concat C=[raw,v5]	0.86/2.57	0.62/1.99	+9.1%/+15.7%	5	48	$<.001/<.001$
Concat C=[raw,v4]	0.86/2.54	0.62/1.94	+8.6%/+17.8%	6	48	$<.001/<.001$
GBM ceiling probe‡	1.11/2.72	0.73/2.07	-51.2%/+5.9%	n/a	n/a	–/–

Table 4: Canonical-split (train 2019–2022/val 2023/test 2024), locked test-2024 headline numbers, every row on identical data. RMSE in °C (std= 4.554°C, `results/canonical_units.json`). p is Diebold–Mariano vs. persistence; both concat rows also differ significantly from the null-control Ridge baseline (see text), toward worse skill. VPT: max consecutive horizon (hours) with NRMSE below 0.4. FSDH: max consecutive horizon (of 48) beating persistence. †Null-control KRR's training set is capped at 3,000 samples; VPT/FSDH not computed for it (never swept over all 48 horizons) – Null-control Ridge and Residual-Ridge instead reuse the equivalent raw-window-Ridge VPT/FSDH sweep from Sec. 6.2. ‡GBM is a predictability-ceiling probe, not an RC-comparison baseline; VPT/FSDH/DM not computed for it.

8 Discussion, Limitations, and Stakeholder Impact

A plain linear Ridge regression on the raw observation window matches or beats every reservoir tested, classical or quantum, on the locked canonical test split. **No quantum-advantage claim**: a 12-qubit density matrix is classically simulable, and the only defensible framing is a per-degree-of-freedom comparison against a dimension-matched classical reservoir, also not beaten (Sec. 6.2). Scope is single-station, temperature-only, no spatial/upstream information. **Real NWP comparison, and it is humbling**: at the only lead times a free public archive offers without running our own NWP pipeline (24h/48h GFS, Open-Meteo's Previous-Runs archive, real KORD 2024 data), GFS skill is +55.5%/+65.3% vs. persistence – far above our best model's +12.9%/+21.4% (null-control Ridge) at the same horizons (p<0.001). No model here is competitive with a real NWP pipeline. **Scope reductions, logged not hidden**: the full ablation planned (2 $\gamma_1 \times$2 multiplexing$\times$3 seeds, 12 drives) was reduced to 2 drives (1 seed, V=1) for time-budget reasons, estimated at ∼21 GPU-hours post-fix and not run. **Real bugs found and fixed, not hidden**: beyond the Tier-1/Tier-2 bugs in Sec. 3.1, a PennyLane auto-queuing double-application bug, a missing-data bug, and a window/raw-sequence index-alignment gap in the canonical-split export were each found and fixed before silently corrupting a result. **Stakeholder impact**: an operational forecaster should, honestly, choose linear regression over this reservoir today; the value here is a validated negative result and mechanistic understanding motivating further work.

9 Conclusion

Dissipation is a real mechanism here, but its signal is redundant with a linear read of the observation history, and the reservoir does not yet beat that baseline, reported with a validated, reproducible stack behind every result.

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